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Progress and recommendations in data ethics governance: a transnational analysis based on data ethics frameworks

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Human society has entered a new stage of development driven by data, where data is gradually becoming a factor of production. Big data and artificial intelligence technologies driven by data are changing the ways of global academic research, social governance, and economic development. In the era of big data, the frequent occurrence of data ethics violations has prompted countries to promote policies in the field of data ethics and to build relatively mature frameworks for data ethics. This study aims to sort out the progress of international data ethics frameworks through online research methods, literature surveys, and content analysis, summarizing the ethical regulations in the rapid development of big data and artificial intelligence. The study employs literature retrieval and search engine retrieval to select ten leading countries or supranational alliances in data and artificial intelligence development and their representative data ethics framework documents for in-depth research, including the United Kingdom (UK), the United States (US), Germany, Australia, Switzerland, Singapore, China, the European Union (EU), the United Nations Educational, Scientific and Cultural Organization (UNESCO), and the Organization for Economic Cooperation and Development (OECD). The study finds that while the specific content of data ethics principles varies among countries, there is a universality in underlying values, with data security and privacy protection comprising the core elements of data ethics governance. Data ethics frameworks in various countries present a synergistic governance model combining legal support, ethical guidance, and regulatory mechanisms, and have included extensive participation from interdisciplinary teams and stakeholders. It is recommended that countries, when improving their data ethics governance systems, integrate their national cultural characteristics based on universal values, draw on wisdom from various academic fields, balance the demands of different stakeholders, and adopt a governance model of legal support—ethical guidance—and regulatory mechanisms, applying diversified governance tools to achieve multi-level collaborative governance.

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Introduction

The exponential growth in data and computational power has fueled the development of data-driven technologies, such as artificial intelligence (AI). With the profound transformation of the data-intensive fourth scientific paradigm and the emergence of open science as a global consensus, human society has entered a new phase of data-driven transformation. The application of data and the development of AI technologies are changing the ways of social governance and economic development, and also bringing new methods and momentum to disciplinary research (Song, 2020). However, in data-rich environments, increasing interconnectedness may challenge accepted social and ethical norms (Mittelstadt and Floridi, 2016). The widespread use of big data, machine learning, and AI raises a series of urgent issues concerning fairness, responsibility, and respect for human rights (Floridi and Taddeo, 2016). In 2018, Facebook was embroiled in a serious data breach scandal, allowing third-party applications to improperly access the personal data of about 87 million users without their consent, which was later illegally sold to Cambridge Analytica and used for targeted political advertising towards American voters, triggering global attention and discussion on data privacy and social media responsibilities. Biomedical big data, due to its sensitivity, has attracted widespread attention. In the utilization of biomedical big data, there are many cases where insurance companies illegally obtain and use personal health data without authorization, deciding on individual insurance coverage based on individual health conditions to maximize their own interests, leading to serious health inequality issues. The Facebook incident is a typical case of a major social public event caused by data breaches, while the abuse of personal health data is a typical phenomenon of individual rights infringement. Data ethics issues are numerous, ranging from personal information leaks to national security impacts, covering a wide range of fields and entities, with a broad impact range and the potential to cause serious social problems.

Data contains immense value and is gradually becoming a strategic asset, bringing about significant social and economic benefits. However, the accompanying ethical issues in data, such as personal privacy leaks, lack of informed consent, exacerbation of data inequality, and challenges on data security, have sparked widespread contemplation: How can we balance the relationship between personal privacy protection and the sharing of data? How can we avoid data divides and data hegemony to maintain data fairness? How can we determine the ownership of data within the entire lifecycle involving multiple stakeholders? The era of big data has adjusted the relationships between people, between people and technology, and between people and society, urgently requiring a moral consensus and the formation of widely accepted ethical norms to address the ethical conflicts and moral dilemmas brought about by the development of big data (Peng, 2020). Data ethics governance refers to the collective efforts of various stakeholders to address ethical challenges arising from data collection, utilization, storage, and related processes (Si and Liu, 2025). This involves managing and regulating these activities through diverse policies and technological tools to tackle issues, such as the digital divide, data misuse, digital identity concerns, and privacy breaches, while simultaneously unlocking their value. The governance process requires multi-stakeholder coordination to achieve consensus throughout its implementation. Yet, we still have doubts about how to regulate the use of data scientifically and systematically, that is, how to conduct data ethics governance? Who are the subjects involved in data ethics governance? What processes should data ethics governance follow? And what governance principles should data ethics adhere to?

The uneven distribution of data values among data subjects has led to conflicts of interest among them. To reconcile the interests of these subjects and to prevent harm while safeguarding individual rights, governments and supranational alliances globally have been actively promoting the formulation of data ethics frameworks, achieving notable progress to date. The data ethics framework provides regulations and guidance on the principles and governance pathways that should be followed in data ethics governance, serving as a significant reference in addressing the issues we have identified. Therefore, to address the numerous challenges in data ethics governance, this study collects several data ethics frameworks with the objective of identifying valuable commonalities from the governance norms of various countries and supranational alliances, which aim to guide a broader spectrum of data ethics governance practices. AI technology, propelled by data, encounters comparable ethical issues, including privacy, ownership, and fairness, which may intensify the risks associated with data ethics violations (Chen, 2020). Consequently, this study also encompasses the analysis of AI ethics governance frameworks.

In the subsequent sections, the author will first delineate the origins of data ethics and highlight notable academic research in this domain. Next, the methodology of this study—specifically the retrieval strategy for data ethics frameworks—will be elucidated. Following this, the paper synthesizes four key findings regarding global advancements in data ethics. Building on global governance experience, four actionable governance recommendations are proposed. Finally, the study concludes with a summary of contributions and an outlook on future developments in the field.

The origin and research status of data ethics

The origin of data ethics. Ethics, as a discipline focused on moral phenomena, constitutes a scientific inquiry into morality (Luo, 1989). The primary task of ethics is to provide a general theoretical framework that norms "what is right and what ought to be done" (Peng, 2020). Data ethics represents a new type of ethical relationship that has emerged in the era of big data, integrating technical and ethical dimensions. It has evolved from computer ethics to internet ethics to information ethics, and essentially is an emerging interdisciplinary field built upon the foundations of computer ethics, internet ethics, and information ethics, representing an inheritance and development of traditional ethics (Song, 2020). The ethical challenges brought by data science can be mapped into the conceptual space defined by the three axes of research: data ethics, algorithm ethics, and practice ethics. Ethical issues can be viewed as points determined by the intersection of multiple axes; thus, addressing data ethics concerns necessitates a comprehensive approach to problem-solving across the entire conceptual landscape.

At present, there is no unified definition of data ethics within the academic community. Luciano Floridi has argued that "data ethics can be defined as a new branch of ethics that studies and evaluates moral problems related to data (including generation, recording, curation, processing, dissemination, sharing and use), algorithms (including AI, artificial agents, machine learning and robots) and corresponding practices (including responsible innovation, programming, hacking and professional codes), in order to formulate and support morally good solutions (e.g., right conducts or right values)" (Floridi and Taddeo, 2016). The *Federal Data Strategy: Data Ethics Framework* issued by United States General Services Administration defines data ethics as "the norms of behavior that promote appropriate judgments and accountability when acquiring, managing, or using data, with the goals of protecting civil liberties, minimizing risks to individuals and society, and maximizing the public good" (US General Services Administration, 2020).

Current state of data ethics research. Over the past decade, scholars have conducted research on data ethics issues, principles of data ethics governance, and the definition of data ethics internationally. Davis and Patterson (2012) authored the internationally recognized book *Ethics of Big Data*, which provides an analysis and critique of ethical issues, including identity, privacy, ownership, and reputation in the era of big data. Qiu et al. (2014) and their colleagues asserted that ethical issues associated with big data technology encompass digital identity, privacy, accessibility, security, and safety, the digital divide, among others. They proposed nine ethical governance principles for big data technology, which include fundamental purposes, responsible research, conflict of interest, respect, privacy, justice, solidarity, transparency, and participation. Jobin et al. (2019) conducted a scoping review of the global landscape of AI ethics guidelines. After analyzing 84 documents on AI ethics principles and guidelines, they identified 11 key ethical values and principles, namely transparency, justice and fairness, non-maleficence, responsibility, privacy, beneficence, freedom and autonomy, trust, dignity, sustainability, and solidarity. Fosch Villaronga and Malgieri (2023) have offered a critical perspective on AI ethics from a more emerging viewpoint. They explore how to "queer" AI ethics to challenge and reassess the normative assumptions and values underlying AI systems. They emphasize the potential of AI technologies to exacerbate discrimination and inequality and propose the necessity of designing more inclusive and equitable AI systems through the lenses of intersectionality and vulnerability.

Floridi and Taddeo (2016) proposed a definition of data ethics. Mittelstadt and Floridi (2016) conducted a comprehensive and systematic review of the academic literature concerning the ethical implications of big data. The article identified five key ethical concern areas: informed consent, privacy, ownership, epistemology, and the "Big Data divide". Furthermore, it proposed six areas that warranted attention but had not yet sparked extensive discussion in the existing literature, including group-level ethics, ethical implications of growing epistemological challenges, effects of big data on fiduciary relationships, the ethics of academic versus commercial practices, ownership of intellectual property derived from big data, and the content of and barriers to meaningful data access rights. Floridi (2018) introduced the concepts of hard ethics and soft ethics. Hard ethics refers to the ethical issues involved in the formulation of new regulations or the challenge of existing ones, focusing on what is morally right or wrong, and what ought or ought not to be done. In contrast, soft ethics is applied subsequent to legal compliance, representing post-compliance ethics, which considers what ought or ought not to be done beyond the confines of existing regulations, rather than opposing, disregarding, or attempting to alter them. Hard ethics provides a moral framework concerning rights, duties, and responsibilities that can influence the formation, shaping, and transformation of laws, while soft ethics focuses on how to better achieve morally good behavior within the legal framework.

Wen et al. (2019) identified three primary factors influencing the ethics of scientific data sharing: inadequate self-regulation by network entities, the misuse of data sharing technologies, and the lack of external regulatory mechanisms. As the social and legal context for data ethics malpractice, external regulatory mechanisms, such as data ethics frameworks and regulations, serve a vital function as both "rule setters" and "guideposts" in navigating the ethical challenges presented by the era of big data. Scholars have conducted research on external regulatory mechanisms, such as data ethics frameworks and policies. For example, Drew (2016) created a data science ethics framework to assist government officials and data scientists in using data science methods within the legal and ethical frameworks, while better protecting the interests of the public. The framework includes six main principles: start with clear user need and public benefit, use data and tools that have the minimum intrusion necessary, create robust data science models, be alert to public perceptions, be as open and accountable as possible, and keep data secure. Si and Liu (2024) conducted an analysis of 78 data ethics policies in the United Kingdom, utilizing grounded theory and social network analysis to explore the UK's data ethics governance framework and stakeholder collaboration network.

External regulatory mechanisms, such as data ethics frameworks, play a crucial role in guiding the governance of data ethics. Consequently, this study aims to deliver a comprehensive analytical review of data ethics frameworks from data-developing countries and supranational alliances. The objective is to summarize global experiences in data ethics governance and provide valuable references for the optimization and enhancement of data ethics governance systems worldwide.

Data ethics framework retrieval strategy

To retrieve data ethics frameworks that are accessible, representative, and comprehensive, this study primarily employs two methods: literature search and search engine retrieval. Given that the research subject focuses on data ethics frameworks rather than literature itself, and considering the extensive scope of retrieval that makes systematic and comprehensive search challenging, this study adopts a non-systematic retrieval approach. The inclusion criteria for data ethics frameworks are as follows: (1) Accessibility—the complete document must be obtainable through conventional retrieval methods; (2) Representativeness—the framework must belong to a nation or organization demonstrating advanced development in big data or AI domains, and the framework should be recognized as a representative document in these fields for that specific nation or organization; (3) Comprehensiveness—the data ethics framework must contain ethical principles with complete structure and substantive content.

Retrieval strategy: (1) Literature search: this method relies on databases, such as Web of Science, PubMed, Scopus, Google Scholar, and others to retrieve articles related to data ethics frameworks. In the Web of Science database, literature was searched using the query TS = ("data ethics" AND "framework"); in the PubMed database, the search was conducted with Title/Abstract = ("data ethics" AND "framework"); in the Scopus database, literature was searched with Title-Abs-Key = ("data ethics" AND "framework"); and in the Google Scholar database, literature and other documents related to "data ethics framework" were searched. By reviewing the titles and abstracts of the literature, appropriate articles were selected for full-text reading, and the names of documents related to data ethics frameworks were collected. Additionally, a snowballing method was employed by reviewing references to further identify literature and gather names of framework documents. (2) Search Engine Query: In mainstream search engines, such as Bing and Google, the keyword "data ethics framework" was utilized to supplement the search for data ethics frameworks.

Ultimately, ten countries and supranational alliances with well-established data ethics frameworks were selected for study. Table 1 presents the representative data ethics frameworks issued by these data-advanced countries and organizations. However, due to the non-systematic and non-comprehensive nature of the retrieval methodology, this study has inherent limitations in its research approach. It cannot guarantee that the included data ethics frameworks fully represent the global panorama of data ethics advancements. Consequently, this research aims to outline the current state of global data ethics governance through a

Table 1 Representative data ethics framework issued by countries and supranational alliances.

Rank	Publication time	Publication agency	Data ethics framework	Data ethics principles
1	2020-09	UK Central Digital And Data Office	Data ethics framework	Transparency; Accountability; Fairness
2	2020-09	US General Services Administration	Federal data strategy: data ethics framework	Uphold applicable statutes, regulations, professional practices, and ethical standards; Respect the public, individuals, and communities; Respect privacy and confidentiality; Act with honesty, integrity, and humility; Hold oneself and others accountable; Promote transparency; Stay informed of developments in the fields of data management and data science
3	2019-10	German Federal Government Data Ethics Commission	Opinion of the Data Ethics Commission	Human dignity; Self-determination; Privacy; Security; Democracy; Justice and Solidarity; Sustainability
4	2019-11	Australian Department of Industry, Science and Resources	Australia's Artificial Intelligence Ethics Framework	Human, societal and environmental wellbeing; Human-centered values; Fairness; Privacy protection and security; Reliability and safety; Transparency and explainability; Contestability; Accountability
5	2018-05	Swiss Personalized Health Network	Ethical Framework for Responsible Data Processing in Personalized Health Research	Respect for persons; Privacy; Data fairness; Accountability
6	2020-01	Personal Data Protection Commission Singapore	Model Artificial Intelligence Governance Framework (second edition)	Explainable; Transparent; Fair; Human-centric; Well-being; Safety
7	2023-03	Chinese Artificial Intelligence Standardization Overall Group	Ethical Governance of Artificial Intelligence Standardization Guidelines 2023	For Human; Sustainability; Collaboration; Privacy; Fairness; Share; Security; Safety; Transparency; Accountability
8	2019-04	European Commission High-Level Expert Group on Artificial Intelligence	Ethics Guidelines for Trustworthy AI	Human agency and oversight; Technical robustness and safety; Privacy and data governance; Transparency; Diversity, non-discrimination and fairness; Societal and environmental wellbeing; Accountability
9	2021-11	United Nations Educational, Scientific and Cultural Organization	Recommendation on the Ethics of Artificial Intelligence	Proportionality and Do No Harm; Safety and security; Fairness and non-discrimination; Sustainability; Right to Privacy, and Data Protection; Human oversight and determination; Transparency and explainability; Responsibility and accountability; Awareness and literacy; Multi-stakeholder and adaptive governance and collaboration
10	2019-05	Organization for Economic Co-operation and Development	OECD AI Principles Overview	Inclusive growth, sustainable development and well-being; Human rights and democratic values, including fairness and privacy; Transparency and explainability; Robustness, security and safety; Accountability

limited perspective, using this lens to analyze and evaluate the progress of international data ethics frameworks. By summarizing exemplary practices in data ethics governance, this study seeks to offer reference and insights for other nations in formulating or refining their ethical frameworks.

Progress in international data ethics governance

Data ethics principles: diversity in specific content and universality of core values. In the era of digital intelligence, governments and supranational alliances have accelerated the exploration and issuance of data ethics frameworks and AI ethical guidelines. Taking into account the variations in political, economic, cultural, and social contexts, there is a diversity in the formulation of data ethics principles among different countries. Nevertheless, some universal core values can be identified, including human-centricity, privacy protection, security, transparency, accountability, and fairness, which form the foundation of global data ethics.

The *Data Ethics Framework* (UK Central Digital and Data Office, 2020) outlines the overarching principles that data-related actions and projects should adhere to, including transparency, accountability, and fairness, concisely summarizing the basic ethical guidelines of data-related practices.

The *Federal Data Strategy: Data Ethics Framework* (US General Services Administration, 2020) embodies core values of human-centricity, privacy protection, transparency, and accountability, while also highlighting the necessity of adhering to the legal landscape, fostering ethical conduct, and keeping abreast of advancements in data science.

The *Opinion of the Data Ethics Commission* (German Federal Government Data Ethics Commission, 2019) identifies "human dignity; self-determination; privacy; security; democracy; justice and solidarity; sustainability" as seven fundamental ethical and legal principles essential to German society. These principles not only embody human-centricity, privacy protection, security, and fairness but also serve as a warning against the potential manipulation of politics through digital technology, advocating for digital technology services that contribute to sustainable societal development.

The eight ethical principles outlined in *Australia's Artificial Intelligence Ethics Framework* (Australian Department of Industry Science and Resources, 2019) aim to promote safe, reliable, and fair practices in AI. Building upon the aforementioned six core values, Australia also permits individuals to challenge the use and outcomes of AI systems, ensuring that the application of these systems benefits humanity, society, and the environment.

The *Ethical Framework for Responsible Data Processing in Personalized Health Research* (Swiss Personalized Health Network, 2018) proposes "respect for persons; privacy; data fairness; accountability" as four ethical principles that bridge human-centricity, privacy protection, fairness, and accountability, declaring their significance in dealing with health-related data.

The guiding principles proposed by the *Model Artificial Intelligence Governance Framework* (Personal Data Protection Commission Singapore, 2020) not only reflect the core values of transparency, fairness, human-centricity and security, but also incorporate explainability and well-being, in order to enhance user trust and comprehension of AI technology utilization.

The ethical guidelines advocated in the *Ethical Governance of Artificial Intelligence Standardization Guidelines 2023* (Chinese Artificial Intelligence Standardization Overall Group, 2023) encompass the aforementioned six core values. Additionally, China places significant emphasis on the harmonious coexistence and sustainable development of new technologies within the social environment, and seeks to achieve collaborative governance of data and AI through a cooperative and shared approach.

The *Ethics Guidelines for Trustworthy AI* (European Commission High-Level Expert Group on Artificial Intelligence, 2019) posits that trustworthy AI should be lawful, ethical, and robust. In addition to embodying the aforementioned six core values, the European Union (EU) also emphasizes the importance of AI technologies serving the welfare of society and the environment.

The *Recommendation on the Ethics of Artificial Intelligence* (UNESCO, 2021) presents ten AI ethics principles that integrate the aforementioned six core values, advocating that AI methodologies should align with specific legitimate objectives, avoid harm to humanity, society, and the environment, contribute to sustainable social development, enhance public understanding and awareness of AI technology, and balance stakeholder participation throughout the lifecycle of AI systems.

The ethical guidelines outlined in the *OECD AI Principles Overview* (OECD, 2019) encompass the six core values mentioned earlier. It also advocates that AI technologies should serve human well-being and promote the sustainable development of society and the environment.

Different governments and supranational alliances have varying interpretations of data ethics principles. However, the underlying universality of their core values can be identified. This integration of specificity and universality not only caters to the unique needs of different legal, political, social, and cultural contexts but also expresses respect for shared human values and interests. It lays the foundation for global data science communication and collaboration, enabling a better adaptation to the challenges posed by the rapid development of data and AI technologies.

Data security and privacy protection: core elements of global data ethics frameworks. From a micro perspective, countries and organizations, such as the United States, Germany, Australia, Switzerland, Singapore, China, as well as the EU, UNESCO, and the OECD, all advocate for data ethics principles that unanimously emphasize the importance of data security and privacy protection, demonstrating that data security and privacy protection are key components of data ethics governance. Table 2 lists key policies and laws at the national and international levels regarding data security and privacy protection. It is evident that laws on data security and privacy protection occupy a significant portion of the data ethics policy landscape in various countries. From a macro perspective, data security and privacy protection have become central components of the data ethics policy systems in various countries. However, persistent challenges remain in reconciling legislative and enforcement discrepancies rooted in distinct socio-political contexts across nations and alliances, the resolution of which could significantly enhance global data ethics governance efficacy.

The United Kingdom holds the largest share of the data market in Europe and is at the forefront of the world in data innovation development and data ethics governance (Feng et al., 2024). Although the UK data ethics framework does not explicitly designate data security and privacy protection as its ethical principles, it is evident from the UK's digital strategy that data security and privacy protection are critical steps of the country's data ethics governance process. However, following Brexit, the UK has had to establish data transfer mechanisms and agreements separate from the EU, which has posed challenges in cross-border data flows, such as difficulties in coordinating data protection standards with the EU and other countries.

The United States' policy landscape in the field of data ethics is clearly focused on data security and privacy protection. However, interstate legislative inconsistencies and federal-state jurisdictional conflicts impose substantial compliance burdens on enterprises. Additionally, the US adopts a market-led, industry

Table 2 Policies and laws in data security and privacy protection of countries and supranational alliances.

Rank	C/SA	Key policies and laws	Main contents
1	United Kingdom	*Data Protection Act (DPA)*	Regulates data protection principles, citizens' data rights, and complaint channels
		UK Digital Strategy (UK Government, 2022)	Emphasizes data security and intellectual property protection
2	United States	*National Security and Personal Data Protection Act* (US Government, 2019)	Emphasizes the data security and privacy protection of nation and citizens
		Uniform Personal Data Protection Act (Uniform Law Commission, 2021)	Emphasizes the data security and privacy protection of business entities holding personal data
		American Data Privacy and Protection Act (US Government, 2021)	Emphasizes the data security and privacy protection of consumers and businesses
		Federal Data Strategy (US Government, 2019)	Adopts "ethical governance" as its primary guiding principle and underscores the urgent need to protect security, privacy, and confidentiality
3	Germany	*Federal Data Protection Act* (Federal Government, 2021)	Provides protection for personal data and data from public institutions, while detailing the rights of data subjects
		Recommendations on Data and Algorithms	Distinguishes ethical principles between responsible data governance and the responsible use of algorithmic systems, and advocates for separate regulation of personal and non-personal data
		Data Strategy of the Federal German Government (Federal German Government, 2021)	Identifies four key areas of action, one of which is "promoting data innovation and the responsible use of data"
4	Australia	*Freedom of Information Act*	Marks the inception of data ethics regulation and forms the foundational legal framework for data ethics supervision in Australia
		Privacy Act	
		Privacy Legislation Amendment (Enhancing Online Privacy and Other Measures) Bill 2021	Further added regulations for online privacy
5	Switzerland	*Federal Act on Data Protection (FADP)*	Safeguards personal privacy and fundamental rights during data processing; establishes the principles of legality, integrity, and proportionality in data processing
		Federal Act on Human Genetic Testing (HGTA)	Protects individuals' privacy and rights in human genetic testing
		Human Research Act (HRA)	Regulates research activities involving humans, ensures ethical and legal compliance
6	Singapore	*Personal Data Protection Act 2012* (Singapore Government, 2012)	Cornerstone of Singapore's data protection legal framework; focuses on the collection, use, disclosure, and protection of personal data
		Cybersecurity Act (Singapore Government, 2018)	Prevents, manages, and responds to cybersecurity threats and incidents, particularly in the context of critical information infrastructure
		Electronic Transactions Act (Singapore Government, 2010)	Fosters the security and reliability of electronic transactions
7	China	*Cybersecurity Law of the People's Republic of China* (National People's Congress of China, 2016)	Data processing activities should comply with laws and regulations, respect social morals, and adhere to cyber ethics
		Data Security Law of the People's Republic of China (National People's Congress of China, 2021a)	
		Personal Information Protection Law of the People's Republic of China (National People's Congress of China, 2021b)	Embodies data ethics principles, such as informed consent, transparency, openness, security, accountability, and public welfare in its provisions on personal information processing
		Regulations on Network Data Security Administration (Cyberspace Administration of China, 2021)	Provides detailed provisions regarding data processing activities that violate legal regulations and ethical morals
8	European Union	*Data Protection Directive* (European Union, 1995)	Protects the rights related to the processing of personal data and sets regulatory requirements for data security and individual privacy
		General Data Protection Regulation (GDPR) (European Union, 2018)	Emphasizes the protection of personal data rights and establishes an independent regulatory body, the ethical red lines in the era of big data have been clearly delineated
9	OECD	*OECD Guidelines on the Protection of Privacy and Transborder Flows of Personal Data* (OECD, 1980)	Outlines eight fundamental principles in the protection of personal data
			OECD Privacy Guidelines—global minimum standards for privacy and data protection

C/SA Countries/Supranational Alliances.

self-regulation-based regulatory model, which has led to frequent data breaches and other security incidents.

Germany advances responsible data governance through its dual public-private legislative model under the *Federal Data Protection Act* (Federal Government, 2021), though constrained public sector data sharing impacts administrative efficiency. What's more, the proposed differentiated regulation for personal and non-personal data in *Recommendations on Data and Algorithms* faces implementation challenges due to classification ambiguities.

Table 3 Legal support and regulatory mechanism in data ethics framework of countries and supranational alliances.

Rank	C/SA	Legal support	Regulatory mechanism
1	United Kingdom	UK's framework complies with the *UK GDPR* and *DPA 2018* in personal data processing, and ensures that analysis and automated decision-making do not result in discriminatory outcomes as defined by the *Equality Act*.	UK requires data users to develop a comprehensive plan for responsible data use, alongside assessment programs and accountability mechanisms for oversight.
2	United States	American framework draws upon numerous legal standards across various industry sectors.	/
3	Germany	German framework references GDPR data subject rights, emphasizes strengthening existing legal tools for data ethics regulation, and suggests cautious adjustments to conflicting parts of current legal frameworks.	Germany has developed a "Critical Pyramid for the Use of Algorithmic Systems and a Risk-Adaptive Regulatory System," categorizing application systems into five levels of risk and proposing corresponding regulatory measures, along with specific recommendations for data ethics oversight.
4	Switzerland	Participating institutions have to comply with Swiss law, including but not limited to the FADP, the HGTA, and the HRA.	Switzerland provides specific action guidelines for accountability mechanisms, including the transparency and auditability of data processing personnel and procedures, appropriate monitoring, performance evaluation, and regular auditing of data security measures, and assessment of the capabilities of institutions managing human biological materials.
5	European Union	Trustworthy AI requires legality, ethics, and robustness, with legality as the priority. Ethical principles are based on fundamental rights outlined in EU treaties, the EU Charter, and international human rights law.	EU promotes accountability mechanisms to hold responsible parties accountable for outcomes throughout the development, deployment, and use of AI systems. A checklist has also been developed to assess AI trustworthiness.
6	UNESCO	UNESCO's framework follows the spirit of the *United Nations Charter*, the *Universal Declaration of Human Rights*, and the *United Nations Declaration on the Right to Development*.	UNESCO proposes the establishment of appropriate oversight, impact assessment, auditing, and due diligence mechanisms to ensure accountability for AI systems and their impacts throughout the entire lifecycle.

C/SA Countries/Supranational Alliances.

China has established an overarching legal framework prioritizing data security and privacy protection, complemented by multiple AI governance guidelines. Although China emphasizes data security and ethical AI governance, its data policies and industry practices reflect a dynamic interplay between national control and digital sovereignty.

As a pioneer in the field of data protection, the EU has issued numerous policies and regulations related to data protection (Fang and Chen, 2024). The *General Data Protection Regulation (GDPR)* (European Union, 2018) has set ethical boundaries in the big data era and is widely regarded as the gold standard for data protection. But the enforcement of GDPR remains uneven, with corporate actors finding ways to bypass its intended protections.

Data ethics collaborative governance model: integration of legal support, ethical guidance, and regulatory mechanism. Single ethical guidance is insufficient for perfecting the data ethics governance system, as data ethics frameworks across countries generally emphasize the importance of collaborative governance. First, the actions of data subjects must comply with the existing legal framework, and the formulation of data ethics norms requires legal foundations for support. Second, ethical guidance is crucial to ensure that data processing activities align with societal values and moral standards. Finally, regulatory mechanisms are essential for the effective enforcement of legal and ethical principles. Therefore, legal support, ethical guidance, and regulatory mechanisms are all critical to data ethics governance. A collaborative governance model of data ethics, combining legal support, ethical guidance, and regulatory mechanisms, has begun to take shape in the data ethics practices of various countries.

Table 3 shows the legal support and regulatory mechanisms in the data ethics framework of countries and supranational alliances. In the data ethics frameworks of various countries, adherence to existing legal norms and the implementation of legal spirit are universally evident. Meanwhile, countries have established a series of data ethics guidelines based on international law and the spirit of their own legal systems. However, theoretical guidance requires the support of mandatory regulatory and accountability mechanisms for effective implementation, and the importance of regulatory measures has been emphasized in various forms within the data ethics frameworks of different countries. Additionally, the operation of data ethics frameworks requires timely regulation and evaluation. To this end, countries, such as the UK and Germany have established data ethics committees, while the EU has appointed ethics officers to ensure the effective implementation of data ethics frameworks (Li and Zhang, 2024).

However, the implementation of the collaborative governance model faces several contradictions in practice. For instance, current legal frameworks may not adequately address the complex ethical issues posed by emerging technologies, leading to inflexibility in ethical guidance that adheres strictly to existing legal systems. Additionally, the vagueness of ethical guidance may undermine its operational feasibility in practice. Furthermore, the enforcement of regulatory and accountability mechanisms often suffers from unclear delineation of responsible entities. How to effectively implement regulatory responsibilities throughout the lifecycle of data subjects remains a critical issue for countries to address. Therefore, the implementation of a collaborative governance model for data ethics requires not only collaboration among diverse stakeholders but also the clear definition of responsible entities and practical operational guidelines. Moreover, data ethics frameworks must be regularly updated to align with societal developments.

Mechanisms for developing a data ethics framework: extensive engagement of interdisciplinary teams and stakeholders. The sources and usage pathways of data are intricate and diverse, with entities involved in the production, processing, and application of data spanning various industries. Consequently, data ethics

Table 4 Engagement of interdisciplinary teams and stakeholders in data ethics frameworks of countries and supranational alliances.

Rank	C/SA	Engagement of interdisciplinary teams and stakeholders
1	United Kingdom	UK's framework identifies "engaging a diverse range of experts" as one of its specific actions.
2	United States	American framework has established an interdepartmental team for the development of the framework, consisting of 14 leaders from various government departments.
3	Germany	The members of the German Federal Government Data Ethics Commission encompass professionals from multiple sectors, including law, industry, education, media, consumer protection, and information technology.
4	Australia	Australia's framework received over 130 written submissions during its consultation phase. Stakeholder roundtables and targeted consultations were held in Sydney, Melbourne, Brisbane, and Canberra.
5	Switzerland	The Swiss Personalized Health Network has established an Ethical, Legal, and Social Implications advisory group to develop the framework.
6	Singapore	Singapore promotes stakeholder engagement in AI development and decision-making, providing feedback channels for opinions and requests for organizational reviews.
7	European Union	EU's framework received feedback from over 500 contributors during the public consultation phase. It also sees diverse, inclusive design teams, stakeholder engagement, and societal dialog as integral components to trustworthy AI.
8	UNESCO	UNESCO acknowledges that the ethical issues associated with AI are highly intricate, requiring a multi-stakeholder and interdisciplinary dialogue to build consensus.
9	OECD	OECD encourages multidisciplinary and multi-stakeholder engagement in the responsible governance of trustworthy AI.

C/SA Countries/Supranational Alliances.

possesses a multidisciplinary intersectional nature, with its impact broadly covering numerous sectors. Throughout the entire lifecycle of data, the rights of usage are likely to circulate among multiple entities of different natures, inevitably involving numerous stakeholders. In the competition for data value, various parties may engage in actions that contradict legal and ethical standards, highlighting the urgent need for the regulation and guidance of data ethics principles. To achieve universality, a data ethics framework must adequately reconcile the demands of all stakeholders during its formulation and incorporate knowledge from different disciplines to ensure inclusivity and adaptability in varying contexts.

Table 4 demonstrates the engagement of interdisciplinary teams and stakeholders in data ethics frameworks of countries and supranational alliances. It is evident that, when establishing their respective data ethics frameworks, countries have considered this issue and regarded the broad participation of interdisciplinary teams and stakeholders as a crucial component of the data ethics framework.

While interdisciplinary collaboration and multi-stakeholder participation can enhance the inclusiveness and adaptability of data ethics frameworks in theory, practical challenges, such as uneven resource distribution, inconsistent public participation quality, and inefficiencies from conflicts of interest persist. To maintain the effectiveness and fairness of these frameworks amid rapid technological changes, clearer responsibility demarcation and more efficient, equitable collaboration mechanisms are essential.

Valuable recommendations for data ethics governance

Developing nationally distinct data ethics frameworks based on shared values and national conditions. Data ethics frameworks or AI ethics frameworks from various countries reveal that the foundational values of their data ethics principles exhibit universality while simultaneously reflecting the political, legal, economic, social, and historical-cultural contexts of their respective nations. Consequently, the data ethics frameworks developed are not only credible and binding for their citizens but also facilitate international governance of data ethics. Therefore, it is recommended that other countries, in formulating their data ethics frameworks, ground them in the shared values while integrating their own national conditions to establish distinctive data ethics frameworks.

The shared values upheld by the international community can be primarily categorized into human-centricity, privacy protection, security, transparency, accountability, and fairness. The ten data ethics frameworks considered in this paper fully or partially encompass these six core values. In the context of globalization, the shared values of data ethics provide a common language and benchmark for international cooperation. However, each country must also take into account its specific national conditions when formulating its own data ethics framework. While respecting universal values, countries should develop a data ethics framework that reflects their own legal systems, cultural traditions, and social needs. Such a framework will not only better serve the nation's development strategy but also ensure the feasibility and effectiveness of data ethics principles in practical implementation. For instance, Chinese culture venerates the concepts of peaceful coexistence and harmonious symbiosis, recognizing that public interest outweighs individual private gain, and advocating that technological development should serve the well-being of humanity. Therefore, when formulating a data ethics framework in China, it is essential to integrate principles, such as "harmonious coexistence," "upholding public interest," and "technology for good" on the foundation of human-centricity, privacy protection, security, transparency, accountability, and fairness, thereby achieving an ethical convergence of shared values and the distinctive cultural ethos of China.

Emphasizing the collaborative synergy of legal support, ethical guidance, and regulatory mechanisms. The interplay of law, hard ethics, and soft ethics constitutes a dynamic normative system. Law provides the fundamental behavioral norms, hard ethics offers the moral foundation for legal formulation, while soft ethics delivers a deeper moral assessment and guidance within the legal framework. Law and ethics are intricately intertwined, mutually influencing each other, and collectively advancing society towards a more just, equitable, and moral direction. The regulatory mechanism provides practical assurance for the enforcement of legal and ethical principles. The outcomes of the implementation of the regulatory mechanism feedback into the legal system and ethical principles, facilitating the revision and improvement of the legal and ethical framework to adapt to technological advancements and social changes.

An initial international data ethics framework has been formed, characterized by a governance model that integrates

legal support, ethical guidance, and regulatory mechanisms. This model is exemplified by the data ethics frameworks of the United Kingdom, Germany, Switzerland, the European Union, and UNESCO. Countries can draw on this collaborative governance approach while improving their data ethics framework. Firstly, it is essential to base the framework on legal norms to ensure its legality and authority. Secondly, soft ethics provides in-depth moral interpretation and behavioral guidance for data ethics issues within the legal framework, complementing legal provisions and guiding practice. Finally, hard ethics promotes the incorporation of key ethical issues into the legislative process, contributing to the continuous optimization and improvement of the data ethics legal system.

The regulatory mechanism serves as a guarantee for the effective implementation of laws and ethics. It is recommended that countries prioritize the establishment of evaluation, accountability, and regulatory mechanisms during the formulation of data ethics frameworks. Additionally, the establishment of dedicated data ethics committees or other regulatory bodies is advised to oversee the implementation of the data ethics framework and fulfill other regulatory responsibilities. However, it is also necessary to be vigilant about the occurrence of the "Whiplash effect," which refers to the phenomenon of excessive regulation resulting from an overabundance of concern regarding technological risks. An overreaction to the potential harms posed by certain technologies or practices, leading to the implementation of overly stringent measures, may restrict the positive application and development of these technologies (Mittelstadt and Floridi, 2016). Therefore, countries need to engage in ethical foresight for proactive intervention, identifying, analyzing, and planning for potential ethical issues and challenges that may arise in the future. This involves considering the potential ethical implications of technological advancements at an early stage of development, thereby preparing in advance to formulate more reasonable and proportionate regulatory measures to safeguard the interests of individuals and society.

Encouraging broad participation of interdisciplinary teams and stakeholders. The interdisciplinary nature and the broad range of stakeholders involved in data ethics necessitate that a robust data ethics framework be developed through collective brainstorming and comprehensive consideration. Interdisciplinary teams can integrate expertise and skills from various fields, while the extensive participation of stakeholders ensures that the needs and perspectives of different groups are taken into account during the framework development process, thereby enhancing the inclusivity and effectiveness of the framework. Diverse participation contributes to capturing the complexity of data usage and processing from different perspectives, ensuring that ethical frameworks reflect the diversity and broad concerns of society. Furthermore, extensive participation aids in raising public awareness of data ethics issues, enhancing trust and acceptance of ethical frameworks, thereby facilitating their implementation and adherence.

In the process of developing data ethics frameworks or AI ethics frameworks, various countries and supranational alliances have incorporated perspectives from different disciplines and stakeholders through the establishment of cross-departmental teams, advisory groups, roundtable discussions, and public consultations. This approach not only integrates profound insights from diverse academic fields conceptually, but also ensures that a wide range of stakeholders can participate in the process procedurally. However, due to the diversity of cultural and ethical contexts across countries, it may be necessary to establish deliberative mechanisms to adjudicate disagreements among stakeholders from different global regions, a role that can be mediated and facilitated by intergovernmental organizations (Jobin et al., 2019). Moreover, it is essential to create specific oversight bodies with enforceable mandates, such as independent review boards, to carry out data governance and ethical audits of AI models, thereby operationalizing regulatory oversight through institutional entities.

In the process of developing data ethics frameworks, countries can establish interdisciplinary collaborative mechanisms and involve diverse stakeholders to ensure that the framework content widely incorporates insights from various disciplines and balances the demands of all stakeholders, while also guaranteeing the transparency of the framework development process. Once the draft framework is completed, public consultation, workshops, and online platforms can be utilized to enhance public participation and understanding of data ethics issues, ensuring that the ethical framework reflects public concerns and needs. During the implementation phase of the framework, channels should be provided for stakeholders and the public to give feedback and comments, ensuring that the framework can be updated in a timely manner to adapt to new challenges.

Applying diversified governance tools for multi-level collaborative governance. The emergence of data ethics issues is complex, primarily rooted in the weakening of human agency, while the objective reasons include the immaturity of data technologies and their misuse. The social context encompasses the absence of external regulatory mechanisms, such as legal regulations, ethical norms, and social oversight (Dong and Cheng, 2017). The complexity of the causes behind data ethics issues dictates the diversity and multi-layered nature of governance tools and methods. In the process of data ethics governance, countries should adeptly employ a variety of governance tools and methods, approaching governance from different levels and perspectives.

At the normative level, an ethical governance system should be constructed with a mainline of legal regulations—ethical framework—ethical standards. This system will be based on legal regulations, guided by an ethical framework, and grounded in ethical standards for specific implementation, thereby ensuring that data ethical governance has legal backing and rational justification.

At the organizational level, a supervision and accountability mechanism should be established to impose strict accountability on entities that violate legal and ethical norms in data practices, ensuring the effective enforcement of ethical governance standards. Additionally, through educational and promotional activities, it is important to deepen citizens' understanding of data, enhance their data literacy, and clarify their individual data rights and obligations, thereby facilitating data processing activities within the legal and ethical framework and effectively protecting personal information and privacy.

At the technical level, it is essential to actively develop and deploy advanced technologies, including data anonymization, data encryption, blockchain, federated learning, and differential privacy, to ensure data security and personal privacy. By utilizing data traceability and auditing technologies, the precise tracking of data flow and the recording of complete data trajectories can be achieved. The adoption of debiasing algorithms is crucial for identifying and mitigating potential biases and injustices within datasets, thereby ensuring the non-discriminatory nature of algorithms and decision-making systems, which enhances the overall fairness of the decision-making process. It is recommended that countries, in their journey towards data governance, adopt a multi-layered collaborative governance approach,

employing a diverse array of governance tools for comprehensive treatment. This strategy aims to safeguard data security and personal privacy while promoting the healthy and sustainable development of data, laying the foundation for building a more just and transparent digital society.

Conclusion

This study analyzes the data ethics frameworks of ten countries and supranational alliances, aiming to clarify the progress of international data ethics governance and summarize international experiences in data ethics governance, thereby providing references for optimizing and improving the data ethics governance systems of various countries. The research finds that there are common principles of data ethics followed by different countries, with core values including human-centricity, privacy protection, security, transparency, accountability, and fairness. Furthermore, policies and legal arrangements regarding data security and privacy protection occupy a central position in the data ethics governance system. In the process of formulating data ethics frameworks, it is essential to emphasize the collaborative synergy among legal support, ethical guidance, and regulatory mechanisms, encourage broad participation from interdisciplinary teams and stakeholders, and effectively utilize diversified governance tools to achieve multi-level collaborative governance.

However, with the rapid development of big data and AI, the ethical issues they generate are evolving in real time. This study, based on existing data ethics frameworks for analysis, may encounter limitations in timeliness regarding its conclusions and outcomes. Scholars have proposed emerging arguments concerning data and AI ethics, such as: traditional binary thinking (e.g., male/female, yes/no) may lead to AI systems' inability to accurately recognize or accommodate non-binary and transgender individuals; inherent limitations in current datasets regarding human diversity representation may result in fairness biases within AI systems; anti-discrimination laws typically grounded in fixed categories struggle to address novel forms of AI-driven discrimination (e.g., intersectional discrimination); and AI technologies may exacerbate societal pressures on individual identity expression, creating compounded identity dilemmas across virtual and physical realms (Fosch Villaronga and Malgieri, 2023). Future research should engage with more radical or emerging critical perspectives, while national and supranational alliances ought to incorporate these evolving ethical challenges and corresponding solutions into their data ethics frameworks through systematic updates.

Data availability

No datasets were generated or analysed during the current study.

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Author contributions

QIU and HU jointly contributed to the conception of this manuscript. QIU completed the collection and analysis of data and materials, and prepared the original draft. HU revised and reviewed the manuscript, and obtained the funding support for the article. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethical approval

This article does not contain any studies with human participants performed by any of the authors.

Informed consent

Informed consent was not required as the study did not involve human participants.

Additional information

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